

ONE PHYSICAL INTENT

Volt-watt curve + sensing and power-base semantics

USE A: operating limit

$$p \leq f(V), \quad 220 \leq V \leq 250$$

USE B: operating limit

$$p \leq f(V), \quad 220 \leq V \leq 270$$

USE C: tracking equation

$$p = f_6(V), \quad 242 \leq V \leq 248$$

EXACT LINEAR ROWS

Concave cap on this domain
No nonlinear or integer block

EXPLICIT ENCODING CHOICE

Exact graph / MPCC / smoothing
Hull remains a relaxation

DEPENDS ON THE FAMILY

Local C2: affine away from patches
Softplus: retain nonzero tails

Cache key includes the occurrence, relation, effective bounds, encoding and scales.
Every specialization keeps its bound assumptions as constraints.